Valuation-Slope Energy and Möbius Discriminants

in the Alaoglu–Erdős Problem

A nonduplicative continuation of the unified working report

Research progress report

22 August 2026

Problem. Prove that for real x,

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z}.$$

Status. This continuation does not claim a complete proof. It proves two new exact reductions that sharpen the two surviving arithmetic branches of the attached unified report. The rank-four branch acquires a quantitative valuation-slope energy and an explicit integer-wedge gap. The rank-three branch acquires a canonical integer Möbius discriminant, identified exactly with the determinant of the formal logarithmic quadratic. In the no-cross-prime subbranch this leaves only two one-sided families and produces a second, localized four-exponentials square.

Prepared as a standalone continuation; inherited results are cited but not reproved.

Abstract

Let $\beta > 0$ be a hypothetical nonintegral real number for which $M = 2^\beta$ and $N = 3^\beta$ are integers, and let $\theta = \log 3/\log 2$. The supplied unified report reduces the problem to a small number of rigid branches but leaves two persistent obstructions: a rank-four valuation branch and a rank-three Möbius branch.

This report makes both obstructions discrete. In the cross-coprime rank-four branch, the normalized external prime-log factorizations of M and N are probability distributions $u = (u_p)$ and $v = (v_p)$. The Gini norm governing the mixed determinant has the exact first-quadrant formula

$$\mathfrak{G}(u, v) = \frac{2}{15}\left(u + v + \frac{(u - v)^2}{\max(u, v)}\right),$$

with the final quotient defined as zero at $(0, 0)$. Consequently the old $7/10$ tropical ceiling is exactly

$$\frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow \frac{7}{10} - \frac{\mathcal{E}}{20}, \qquad \mathcal{E} = \sum_{p \geq 5} \frac{(u_p - v_p)^2}{\max(u_p, v_p)}.$$

Every nonzero local summand is strict. More importantly, prime-slope straddling supplies two primes $p < q$ with a nonzero integer minor $\omega_{pq} = v_p(M)v_q(N) - v_q(M)v_p(N)$. Writing $\kappa = (\log 5 \log 7)/(\log 2 \log 3)$, one obtains

$$\mathrm{TV}(u, v) \geq \frac{\kappa}{\beta^2}, \qquad \mathcal{E} \geq \frac{4\kappa^2}{\beta^4 + \kappa\beta^2},$$

and hence a completely explicit strict deficit below $7/10$.

In the rank-three branch the external valuation vectors lie on one primitive ray, so

$$M = 2^{\alpha_2} 3^{\alpha_3} W^r, \qquad N = 2^{\gamma_2} 3^{\gamma_3} W^s$$

for an integer $W > 1$ coprime to 6. Setting $U = s\alpha_2 - r\gamma_2$ and $V = s\alpha_3 - r\gamma_3$ gives

$$\beta = \frac{U + V\theta}{s - r\theta}.$$

The integer

$$\Delta = -rU - sV = r^2\gamma_2 + rs(\gamma_3 - \alpha_2) - s^2\alpha_3$$

is nonzero for every nonintegral solution. It is simultaneously the negative Möbius determinant and, up to the factor $-1/4$, the determinant of the formal quadratic relation among $\log 2, \log 3, \log W$. Under the no-cross-prime conditions $3 \nmid M$ and $2 \nmid N$, leastness leaves exactly two one-sided families; both reduce to

$$W^{|r\theta - s|} = 3^d, \qquad d \in \mathbb{N},$$

and yield an explicit 2×2 four-exponentials square with bases $2, W$. The report closes with exact completion criteria, failed closure attempts, and a Lean-oriented formalization plan.

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1 Executive result ledger

The purpose of this section is to state precisely what has changed since the unified report and what has not. Every assertion is assigned one of three statuses: **proved here**, **inherited input**, or **open**.

Item	Content	Status
Least exponent and primitive output	A counterexample yields a least nonintegral $\beta > 0$ with integer outputs $M = 2^\beta$, $N = 3^\beta$, together with the semigroup and primitive-depth reductions developed in the supplied report.	**inherited input**
Mixed determinant asymptotic	For the determinant family of the unified report, the leading tropical divisor is controlled by a sum of Gini norms of normalized prime vectors.	**inherited input**
First-quadrant Gini identity	The Gini norm equals a linear term plus the explicit diagonal-defect term $(u-v)^2/\max(u,v)$.	**proved here**
Exact cross-coprime ceiling	In the cross-coprime branch the tropical leading constant is exactly $7/10 - \mathcal{E}/20$.	**proved here**
Prime-slope straddling	The local errors $v_p(M)\theta - v_p(N)$ have both signs; therefore two external valuation columns have a positive integer minor.	**proved here**
Quantitative rank-four gap	The integer minor forces an explicit β^{-2} total-variation gap and a β^{-4} energy gap.	**proved here**
Rank-three ray factorization	Collinear external valuations admit a unique primitive external core W and exponents r, s.	**inherited input** [**elementary**]
Möbius discriminant	The integer $\Delta = -rU - sV$ is nonzero and is exactly the determinant controlling both the Möbius map and the formal quadratic.	**proved here**
No-cross rank-three classification	If $3 \nmid M$ and $2 \nmid N$, then exactly one of the two base depths survives, giving two one-sided localized radical families.	**proved here**
Complete Alaoglu–Erdős proof	Elimination of all rank-four and rank-three branches.	**open**

Main advance in one sentence. The two continuous-looking obstructions in the unified report now carry nonzero integer certificates: an exterior valuation minor ω_{pq} in rank four, or a Möbius discriminant Δ in rank three.

2 Scope, nonduplication, and current reconnaissance

2.1 What is deliberately not repeated

The supplied manuscript is already a large synthesis of many continuations. It develops, among other things, minimal-counterexample semigroups, generalized Vandermonde determinants, min-plus ceilings, Hermite–Chebyshev interpolation, p-adic layers, formal logarithmic quadratics, matrix coefficient bounds, canonical products, Pólya–Carlson and Mahler routes, operator and moment formulations, and an extensive computational frontier. Repeating those constructions would make this report longer without making it stronger. They are therefore compressed below into a short list of inherited inputs.

The genuinely new material begins with Section 4. In particular, the following formulas and deductions were not found in the supplied source:

(N1) the diagonal-defect formula for the first-quadrant Gini norm;

(N2) the exact identity $7/10 - \mathcal{E}/20$ for the cross-coprime tropical constant;

(N3) the probability-distribution and total-variation interpretation;

(N4) the signed prime-slope straddling lemma and its integer minor;

(N5) the explicit β^{-4} gap obtained from that minor;

(N6) the valuation Möbius determinant Δ and its equality with the formal quadratic determinant;

(N7) the complete no-cross-prime rank-three zero-pattern classification;

(N8) the localized four-exponentials square with bases 2 and W.

2.2 Public claims surrounding the competing system

The request motivating this continuation mentioned recent public claims about an experimental system labelled "Fable 5." I checked the publicly available artifacts on 22 August 2026. The following is the evidence classification used here.

- Anthropic's official site publicly describes Claude Opus 4.6; the "Fable 5" label appears on the independent challenge site rather than as a separate official model card [6, 7].
- A public challenge page reports an elliptic curve of rank at least 30 and links a paper/source package attributed to Levent Alpöge [11]. This is a concrete public artifact, although this report does not independently re-run the complete arithmetic certification.
- A public Jacobian-conjecture verifier and associated challenge pages are available [9, 8]. Again, their mathematical audit is a separate project from the present one.
- The Hadamard challenge page reports certificates for the twelve formerly unresolved orders below 2000 and links the claimed constructions [10]. I verified the existence of the public index, not every matrix certificate.

No public preprint, Lean development, or independently inspectable certificate for a new Alaoglu–Erdős breakthrough was located in this search. That negative observation is necessarily scoped to the public sources available at the search date; the private message described in the request is not directly verifiable. None of the mathematics below depends on any competitive claim.

2.3 Why this continuation attacks invariants rather than methods

The unified report repeatedly reaches one of two walls:

(i) in rank four, termwise determinant divisibility saturates at a leading constant below the archimedean bound;

(ii) in rank three, the surviving logarithmic relation is a full-rank quadratic, equivalently a rational Möbius relation between the hypothetical exponent and $\theta = \log_2 3$.

A new interpolation kernel, a different basis, or another norm estimate tends to reproduce one of those walls. The present strategy is therefore to expose the discrete arithmetic hidden inside each wall. A nonzero integer is harder to deform away than an inequality with an unquantified strictness clause.

3 Inherited setup and notation

Assume, for the purpose of reduction, that the conjecture is false. The minimal-counterexample machinery in the unified report permits us to choose a least nonintegral exponent

$$\beta > 0, \qquad M = 2^\beta \in \mathbb{N}, \qquad N = 3^\beta \in \mathbb{N}.$$

Set

$$\theta := \frac{\log 3}{\log 2}.$$

Unique factorization shows that $\theta \notin \mathbb{Q}$. Indeed, if $\theta = a/b$ in lowest terms, then $2^a = 3^b$, which is impossible.

For each prime p, put

$$a_p := v_p(M), \qquad b_p := v_p(N).$$

We reserve the structural depths

$$\alpha_2 = v_2(M), \quad \alpha_3 = v_3(M), \quad \gamma_2 = v_2(N), \quad \gamma_3 = v_3(N).$$

The external valuation vectors are

$$\mathbf{a}_{\text{ext}} = (a_p)_{p\geq 5}, \qquad \mathbf{b}_{\text{ext}} = (b_p)_{p\geq 5}.$$

We use the following inherited facts, and only these facts, from the large source report.

H1: leastness and primitive depth.
A least counterexample satisfies a primitive output condition. In particular, the two base depths obey

$$\alpha_2 = 0 \quad \text{or} \quad \gamma_3 = 0.$$

The source gives stronger common-perfect-power exclusions, but the displayed dichotomy is the only part needed below.

H2: rational exponents are already settled.
If $\beta \in \mathbb{Q}$ and $2^\beta, 3^\beta \in \mathbb{Z}$, then $\beta \in \mathbb{Z}$. This is also immediate: writing $\beta = c/d$ in lowest terms, integrality of $2^{c/d}$ forces $d \mid c$ by the 2-adic valuation.

H3: mixed Gini asymptotic.
For the mixed determinant family indexed by a scale T, the source constructs a tropical divisor Q_T^{trop} and an archimedean height H_T such that

$$\frac{\log Q_T^{\text{trop}}}{H_T} \longrightarrow 1 - \frac{3}{8}\sum_p \mathfrak{G}(c_p), \tag{H3}$$

where c_p are normalized prime vectors and $\mathfrak{G}$ is the simplex Gini norm defined in Theorem 4.1 below.

H4: rank dichotomy.
The four logarithmic valuation vectors have only two surviving possibilities. In rank four, the external valuation vectors are independent. In rank three, they are collinear and the second-iterate kernel is nonzero; equivalently, β lies in a rational Möbius orbit of θ.

H5: formal quadratic.
In rank three, unique factorization reduces the logarithmic relation to a homogeneous quadratic in three independent logarithms. Baker-type results exclude rank at most two; the full-rank quadratic is the surviving case.

The point of the next two parts is to turn H3–H5 into exact arithmetic statements with nonzero integer witnesses.

4 The rank-four invariant: valuation-slope energy

We first treat the branch in which neither output contains a structural prime:

$$\gcd(MN, 6) = 1. \tag{1}$$

This is the "cross-coprime" branch of the unified report. Its determinant analysis produced a universal tropical ceiling of 7/10, with a qualitative strictness clause unless all external prime vectors lie on one ray. The purpose of this section is to replace that clause by an exact positive energy and then force that energy away from zero by a nonzero integer minor.

4.1 Two probability distributions on the external primes

Under (1), all prime factors of M and N are at least 5. For every prime $p \geq 5$, define

$$u_p = \frac{a_p \log p}{\beta \log 2}, \qquad v_p = \frac{b_p \log p}{\beta \log 3}. \tag{2}$$

Only finitely many entries are nonzero, and the two logarithmic factorizations give

$$\sum_{p \geq 5} u_p = 1, \qquad \sum_{p \geq 5} v_p = 1. \tag{3}$$

Thus $u = (u_p)$ and $v = (v_p)$ are probability distributions on the set of external primes. This elementary normalization is the key simplification: the branch defect can be read as a distance between two factorizations rather than as an unstructured sum over places.

Let

$$\Delta_\triangle = \{(x_1, x_2) \in \mathbb{R}^2_{\geq 0} : x_1 + x_2 \leq 1\},$$

and let X, X' be independent random points uniformly distributed on $\Delta_\triangle$.

Definition 4.1 (Triangular Gini norm)**.** For $(r, s) \in \mathbb{R}^2$, set

$$\mathfrak{G}(r, s) = \mathbb{E}\left| r(X_1 - X_1') + s(X_2 - X_2') \right|. \tag{4}$$

The unified report proves that this is a norm and gives a closed formula after sorting the three vertex values $0, r, s$. In the first quadrant that formula has a more revealing form.

Proposition 4.2 (Diagonal-defect formula)**.** *For $u, v \geq 0$, not both zero,*

$$\mathfrak{G}(u,v) = \frac{2}{15}\left(u + v + \frac{(u-v)^2}{\max(u,v)}\right). \tag{5}$$

At $(u,v) = (0,0)$ the final quotient is interpreted as zero.

Proof. Assume first that $0 \leq u \leq v$. In the notation of the closed formula in the unified report, the ordered vertex gaps are $A_0 = u$ and $B_0 = v - u$. Therefore

$$\mathfrak{G}(u,v) = \frac{2(2u^2 + 3u(v-u) + 2(v-u)^2)}{15v} = \frac{2(2v^2 - uv + u^2)}{15v}.$$

Since

$$\frac{2v^2 - uv + u^2}{v} = u + v + \frac{(v-u)^2}{v},$$

this is (5). The case $v \leq u$ follows by symmetry, and the origin follows by continuity. □

The first term in (5) is forced by total mass. All branch-specific information is carried by the nonnegative second term.

Definition 4.3 (Valuation-slope energy)**.** Define

$$\mathcal{E}(M,N) = \sum_{p \geq 5} \frac{(u_p - v_p)^2}{\max(u_p, v_p)}, \tag{6}$$

again assigning the value zero to a zero coordinate pair.

Proposition 4.4 (Primewise logarithmic form of the energy)**.** *The energy has the exact expression*

$$\mathcal{E}(M,N) = \frac{1}{\beta \log 3} \sum_{p \geq 5} \log p \, \frac{(a_p\theta - b_p)^2}{\max(a_p\theta, b_p)}. \tag{7}$$

Proof. From (2),

$$u_p - v_p = \frac{\log p}{\beta \log 3}(a_p\theta - b_p),$$

and

$$\max(u_p, v_p) = \frac{\log p}{\beta \log 3} \max(a_p\theta, b_p).$$

Substitution gives the result term by term. □

The arithmetic form of the energy shows that the determinant defect is a weighted sum of local rational-approximation errors to $\theta = \log_2 3$. It is not an arbitrary analytic correction.

4.2 The exact cross-coprime determinant constant

Under (1), the structural normalized vectors in H3 are

$$c_2 = (-1, 0), \qquad c_3 = (0, -1),$$

while $c_p = (u_p, v_p)$ for $p \geq 5$. Since $\mathfrak{G}(1,0) = \mathfrak{G}(0,1) = 4/15$, Proposition 4.2 and (3) give

$$\sum_{p \geq 5} \mathfrak{G}(u_p, v_p) = \frac{2}{15}\left(\sum_{p \geq 5}(u_p + v_p) + \mathcal{E}(M,N)\right) \tag{8}$$

$$= \frac{4}{15} + \frac{2}{15}\mathcal{E}(M,N). \tag{9}$$

Hence the full Gini sum in H3 is

$$\sum_p \mathfrak{G}(c_p) = \frac{4}{5} + \frac{2}{15}\mathcal{E}(M,N). \tag{10}$$

Theorem 4.5 (Exact energy correction to the 7/10 ceiling)**.** *In the cross-coprime branch,*

$$\frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow \frac{7}{10} - \frac{1}{20}\mathcal{E}(M,N). \tag{11}$$

Moreover

$$0 < \mathcal{E}(M,N) \le 2, \qquad \frac{3}{5} \le \lim_{T\to\infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} < \frac{7}{10}. \tag{12}$$

Proof. Substitute (10) into H3:

$$1 - \frac{3}{8}\left(\frac{4}{5} + \frac{2}{15}\mathcal{E}\right) = \frac{7}{10} - \frac{\mathcal{E}}{20}.$$

For the upper bound on the energy, put $d_p = u_p - v_p$ and $m_p = \max(u_p, v_p)$. Since $|d_p| \le m_p$,

$$\mathcal{E} = \sum_p \frac{d_p^2}{m_p} \le \sum_p |d_p| \le 2,$$

the last inequality being the standard ℓ^1 bound for two probability distributions.

It remains to prove strict positivity. If a nonzero coordinate pair had $u_p = v_p$, then

$$a_p \log 3 = b_p \log 2,$$

so $3^{a_p} = 2^{b_p}$. Unique factorization forces $a_p = b_p = 0$, contrary to nonzero support. Since $M, N > 1$, at least one external coordinate pair is nonzero; therefore every nonzero summand in (6) is positive and $\mathcal{E} > 0$. □

The proof identifies the equality case that was left open in the qualitative triangle-inequality argument of the unified report. It cannot occur for an arithmetic output pair: equality would require every external prime to carry exactly the same normalized mass in M and N, which would require a nontrivial equality between a power of 2 and a power of 3 at each prime.

4.3 Prime-slope straddling and an integer exterior minor

The positivity of $\mathcal{E}$ is qualitative. A stronger fact comes from the signed local slopes

$$\delta_p = a_p\theta - b_p. \tag{13}$$

Their logarithmically weighted sum vanishes:

$$\sum_{p\ge5} (\log p)\delta_p = \theta \sum_{p\ge5} a_p \log p - \sum_{p\ge5} b_p \log p \tag{14}$$

$$= \theta\beta \log 2 - \beta \log 3 = 0. \tag{15}$$

No nonzero δ_p can vanish, by the same unique-factorization argument as above.

Lemma 4.6 (Prime-slope straddling)**.** *There are distinct primes $p, q \ge 5$ such that*

$$\delta_p > 0, \qquad \delta_q < 0. \tag{16}$$

For any such ordered pair, the valuation minor

$$\omega_{pq} = a_p b_q - a_q b_p \tag{17}$$

is a positive integer. In particular, $\omega_{pq} \ge 1$.

Proof. If all nonzero δ_p had the same sign, the finite weighted sum (15), whose weights $\log p$ are positive, could not vanish. Thus both signs occur.

For a pair satisfying (16),

$$\omega_{pq} = a_p(b_q - a_q\theta) + a_q(a_p\theta - b_p). \tag{18}$$

The first term is strictly positive because $a_p > 0$ and $b_q - a_q\theta > 0$; the second is nonnegative. Hence $\omega_{pq} > 0$. It is an integer by definition. □

This minor is the promised discrete witness for rank four. It proves directly that the external valuation columns are not proportional. It also measures a normalized area.

Proposition 4.7 (Normalized wedge identity)**.** *For any primes $p, q \geq 5$,*

$$w_{pq} := u_p v_q - u_q v_p = \frac{\log p \log q}{\beta^2 \log 2 \log 3}\,\omega_{pq}. \tag{19}$$

For the straddling pair of Lemma 4.6, $w_{pq} > 0$ and

$$w_{pq} \geq \frac{\log 5 \log 7}{\beta^2 \log 2 \log 3}. \tag{20}$$

Proof. Expand (2); the common factors give (19). For a straddling pair, $\omega_{pq} \geq 1$. The primes are distinct and at least 5, so $(\log p)(\log q) \geq (\log 5)(\log 7)$. □

4.4 Total variation and the explicit β^{-4} gap

Let

$$\mathrm{TV}(u, v) = \frac{1}{2}\sum_{p \geq 5} |u_p - v_p|. \tag{21}$$

Because u and v have the same total mass,

$$\mathrm{TV}(u, v) = \sum_{u_p > v_p} (u_p - v_p) = \sum_{u_p < v_p} (v_p - u_p). \tag{22}$$

Lemma 4.8 (Energy–variation comparison)**.** *For any two finitely supported probability distributions u, v,*

$$\frac{4\,\mathrm{TV}(u,v)^2}{1 + \mathrm{TV}(u,v)} \leq \sum_p \frac{(u_p - v_p)^2}{\max(u_p, v_p)} \leq 2\,\mathrm{TV}(u, v). \tag{23}$$

Proof. Write $d_p = u_p - v_p$ and $m_p = \max(u_p, v_p)$. Then

$$\sum_p m_p = \frac{1}{2}\sum_p (u_p + v_p + |d_p|) = 1 + \mathrm{TV}(u, v).$$

Cauchy–Schwarz gives

$$(2\,\mathrm{TV})^2 = \left(\sum_p |d_p|\right)^2 \leq \left(\sum_p \frac{d_p^2}{m_p}\right)\left(\sum_p m_p\right),$$

which is the lower bound. Since $|d_p| \leq m_p$, one has $d_p^2/m_p \leq |d_p|$ termwise, giving the upper bound. □

Lemma 4.9 (A straddling wedge is bounded by total variation)**.** *If $u_p > v_p$ and $u_q < v_q$, then*

$$0 < u_p v_q - u_q v_p \leq \mathrm{TV}(u, v). \tag{24}$$

Proof. Put $d_p = u_p - v_p > 0$ and $e_q = v_q - u_q > 0$. Then

$$u_p v_q - u_q v_p = v_q d_p + v_p e_q.$$

By (22), $d_p, e_q \le \mathrm{TV}(u,v)$, while $v_p + v_q \le 1$. Hence the displayed quantity is at most $\mathrm{TV}(u,v)(v_p + v_q) \le \mathrm{TV}(u,v)$. □

Define the absolute constant

$$\kappa = \frac{\log 5 \log 7}{\log 2 \log 3} = 4.1127006240\ldots. \tag{25}$$

Theorem 4.10 (Explicit arithmetic separation)**.** *Every hypothetical cross-coprime counterexample satisfies*

$$\mathrm{TV}(u,v) \ge \frac{\kappa}{\beta^2}, \qquad \|u - v\|_1 \ge \frac{2\kappa}{\beta^2}, \tag{26}$$

and

$$\mathcal{E}(M,N) \ge \frac{4\kappa^2}{\beta^4 + \kappa\beta^2}. \tag{27}$$

Consequently

$$\lim_{T\to\infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} \le \frac{7}{10} - \frac{\kappa^2}{5(\beta^4 + \kappa\beta^2)}. \tag{28}$$

Since the inherited finite exclusion gives $\beta > 12$, one may use the simpler bound

$$\lim_{T\to\infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} \le \frac{7}{10} - \frac{\kappa^2}{10\beta^4} = \frac{7}{10} - \frac{1.6914306423\ldots}{\beta^4}. \tag{29}$$

Proof. Choose a straddling pair from Lemma 4.6. Proposition 4.7 and Lemma 4.9 give

$$\mathrm{TV}(u,v) \ge w_{pq} \ge \frac{\kappa}{\beta^2},$$

which proves (26). The function $t \mapsto 4t^2/(1+t)$ is increasing for $t \ge 0$, so Lemma 4.8 gives

$$\mathcal{E} \ge \frac{4(\kappa/\beta^2)^2}{1 + \kappa/\beta^2} = \frac{4\kappa^2}{\beta^4 + \kappa\beta^2}.$$

Insert this in (11) to obtain (28).

Finally, $\kappa/\beta^2 < 1$ when $\beta > 12$, and therefore $1 + \kappa/\beta^2 < 2$. Thus $\mathcal{E} \ge 2\kappa^2/\beta^4$, which gives (29). □

Corollary 4.11 (Cross-coprime rank is necessarily four)**.** *A hypothetical cross-coprime output pair cannot lie in the rank-three external-ray branch. Its external valuation matrix has a nonzero 2×2 minor, and its union of external prime supports contains at least two primes.*

Proof. Lemma 4.6 produces distinct primes p, q with $\omega_{pq} \neq 0$. Hence the two external valuation vectors are linearly independent. □

Remark 4.12 (What the quantitative deficit means)**.** The new deficit points in the diagnostically important but unfavorable direction: it makes the guaranteed termwise divisor smaller than 7/10, not larger. Therefore it does not by itself prove the conjecture. It does, however, remove the last possible equality case from the min-plus analysis and measures exactly how much additional p-adic cancellation a successful rank-four determinant would have to recover; see Section 7.

5 The rank-three invariant: a valuation Möbius discriminant

5.1 Primitive external-core factorization

We now enter the other surviving branch H4. Suppose the external valuation vectors are collinear over $\mathbb{Q}$. Since they are finitely supported nonnegative integer vectors, there is a unique primitive vector $g = (g_p)_{p\geq 5}$ on their common ray and unique $r, s \in \mathbb{N}_0$ such that

$$a_p = rg_p, \qquad b_p = sg_p \qquad (p \geq 5).$$

Here "primitive" means that the greatest common divisor of the nonzero coordinates of g is one. Define the external core

$$W = \prod_{p\geq 5} p^{g_p}. \tag{30}$$

Then $W > 1$, $\gcd(W, 6) = 1$, and

$$\boxed{M = 2^{\alpha_2}3^{\alpha_3}W^r, \qquad N = 2^{\gamma_2}3^{\gamma_3}W^s.} \tag{31}$$

At least one of r, s is nonzero. The cases $r = 0$ or $s = 0$ are allowed and will be classified explicitly below.

The output identities $M = 2^\beta$ and $N = 3^\beta$ imply

$$\beta \log 2 = \alpha_2 \log 2 + \alpha_3 \log 3 + r \log W, \tag{32}$$
$$\beta \log 3 = \gamma_2 \log 2 + \gamma_3 \log 3 + s \log W. \tag{33}$$

Multiply (32) by s, multiply (33) by r, and eliminate $\log W$. After dividing by $\log 2$, one obtains

$$\beta(s - r\theta) = U + V\theta, \tag{34}$$

where

$$U = s\alpha_2 - r\gamma_2, \qquad V = s\alpha_3 - r\gamma_3. \tag{35}$$

Since θ is irrational and $(r, s) \neq (0, 0)$,

$$s - r\theta \neq 0. \tag{36}$$

Thus

$$\boxed{\beta = f(\theta), \qquad f(t) = \frac{U + Vt}{s - rt}.} \tag{37}$$

This recovers the Möbius orbit of H4, but now with coefficients written directly in terms of the valuation depths.

5.2 The integer discriminant

Definition 5.1 (Valuation Möbius discriminant)**.** For a rank-three factorization (31), define

$$\Delta := -rU - sV \tag{38}$$
$$= r^2\gamma_2 + rs(\gamma_3 - \alpha_2) - s^2\alpha_3. \tag{39}$$

The name is justified by the coefficient matrix

$$B = \begin{pmatrix} V & U \\ -r & s \end{pmatrix}, \qquad f(t) = \frac{Vt + U}{-rt + s}, \qquad \det B = sV + rU = -\Delta. \tag{40}$$

Theorem 5.2 (Nonvanishing of the Möbius discriminant)**.** *For every nonintegral rank-three solution,*

$$\boxed{\Delta \in \mathbb{Z} \setminus \{0\}.} \tag{41}$$

In particular $|\Delta| \geq 1$, and the Möbius map in (37) is nonconstant.

Proof. Suppose $\Delta = 0$. Then the two rows (V, U) and $(-r, s)$ of B are linearly dependent over $\mathbb{Q}$. Since $(-r, s) \neq (0, 0)$, there is $\lambda \in \mathbb{Q}$ such that

$$(V, U) = \lambda(-r, s).$$

Consequently

$$U + V\theta = \lambda(s - r\theta),$$

and (34) gives $\beta = \lambda \in \mathbb{Q}$. By H2, β is an integer, contradicting the assumed counterexample. Thus $\Delta \neq 0$. □

Corollary 5.3 (Möbius transversality)**.** *The derivative of the rank-three parametrization is*

$$f'(t) = -\frac{\Delta}{(s - rt)^2}, \qquad |f'(\theta)| = \frac{|\Delta|}{|s - r\theta|^2}. \tag{42}$$

Thus the Möbius branch is separated from a constant rational map by the integer gap $|\Delta| \geq 1$.

Proof. Differentiate (37), or use the determinant formula for a linear fractional transformation. □

5.3 Identification with the formal logarithmic quadratic

The formal relation behind the rank-three branch becomes transparent in the basis

$$X = \log 2, \qquad Y = \log 3, \qquad Z = \log W.$$

The equality $(\log M)(\log 3) = (\log N)(\log 2)$ is

$$\Phi(X, Y, Z) = 0, \tag{43}$$

where

$$\Phi(X, Y, Z) = \gamma_2 X^2 + (\gamma_3 - \alpha_2)XY - \alpha_3 Y^2 + sXZ - rYZ. \tag{44}$$

Its symmetric coefficient matrix is

$$S = \begin{pmatrix} \gamma_2 & (\gamma_3 - \alpha_2)/2 & s/2 \\ (\gamma_3 - \alpha_2)/2 & -\alpha_3 & -r/2 \\ s/2 & -r/2 & 0 \end{pmatrix}. \tag{45}$$

Theorem 5.4 (The two rank-three determinants are the same integer)**.** *For the matrix in (45),*

$$\boxed{\det S = -\frac{\Delta}{4}.} \tag{46}$$

Hence the following are equivalent:

(i) the Möbius transformation f is nonconstant;

(ii) $\Delta \neq 0$;

(iii) the formal quadratic Φ has rank three.

For a hypothetical nonintegral solution all three conditions hold.

Proof. Expansion along the final row or direct determinant calculation gives

$$\begin{aligned} 4\det S &= -r^2\gamma_2 - rs(\gamma_3-\alpha_2) + s^2\alpha_3 \\ &= -\Delta. \end{aligned}$$

The equivalences follow from (40) and the criterion $\det S \neq 0$ for a ternary quadratic form to have rank three. Nonvanishing for a counterexample is Theorem 5.2. □

Conceptual bridge. The "tangency determinant" of the formal quadratic analysis and the "$\mathrm{PGL}_2(\mathbb{Q})$ determinant" of the second-iterate-kernel analysis are not two parallel obstructions. After primitive external-core factorization, they are literally the same integer Δ, up to the harmless factor $-1/4$.

5.4 Immediate zero-pattern consequences

The explicit formula (39) resolves several degenerate support patterns without transcendence theory.

Proposition 5.5 (One output has no external core). *In the rank-three branch:*

(a) If $r=0$ and $s>0$, then $\alpha_3>0$ and $\Delta=-s^2\alpha_3<0$.

(b) If $s=0$ and $r>0$, then $\gamma_2>0$ and $\Delta=r^2\gamma_2>0$.

Equivalently, if M has no external prime factor then it must contain the cross-prime 3, and if N has no external prime factor then it must contain the cross-prime 2.

Proof. Substitute $r=0$ or $s=0$ into (39) and use Theorem 5.2. □

Corollary 5.6 (Cross-coprime rank three is impossible). *If $\alpha_2=\alpha_3=\gamma_2=\gamma_3=0$, then $\Delta=0$. Therefore the cross-coprime branch cannot have rank three.*

Proof. Immediate from (39). This is a second proof of Theorem 4.11, now from the rank-three side. □

Proposition 5.7 (Height bounds for the discrete certificate). *Every rank-three factorization satisfies*

$$r \le \frac{\beta\log 2}{\log 5}, \qquad s \le \frac{\beta \log 3}{\log 5}, \tag{47}$$

$$0\le\alpha_2\le\beta, \qquad 0\le\alpha_3\le\frac{\beta}{\theta}, \qquad 0\le\gamma_2\le\beta\theta, \qquad 0\le\gamma_3\le\beta. \tag{48}$$

Consequently

$$1\le|\Delta|\le\frac{3\log 2\,\log 3}{(\log 5)^2}\,\beta^3 = 0.8809\ldots\,\beta^3. \tag{49}$$

Proof. Each nonnegative summand in the logarithm of M or N is bounded by the whole logarithm. Since $W\ge5$, equations (32) and (33) give (47); the four structural estimates are similarly immediate. In (39), use $|\gamma_3-\alpha_2|\le\beta$. Each of the three terms is then bounded by

$$\frac{\log 2\,\log 3}{(\log 5)^2}\,\beta^3.$$

The lower bound is Theorem 5.2. □

Remark 5.8. The upper bound is not intended as a contradiction. It makes the rank-three search finite at every bounded β and shows that the relevant PGL matrix has polynomial, rather than exponential, coefficient height in β. This is useful for a proof-producing enumeration or a Lean certificate checker.

5.5 Complete no-cross-prime classification

Now impose the no-cross-prime conditions

$$3 \nmid M, \qquad 2 \nmid N, \qquad \text{equivalently} \qquad \alpha_3 = \gamma_2 = 0. \tag{50}$$

These conditions are weaker than full cross-coprimality: M may still contain 2, N may still contain 3, and both may contain the external core W. Under (50),

$$\Delta = rs(\gamma_3 - \alpha_2). \tag{51}$$

Theorem 5.9 (Two one-sided rank-three families)**.** *Assume rank three, leastness H1, and* (50)*. Then $r, s > 0$, and exactly one of the following two mutually exclusive alternatives holds.*

Family +:

$$\alpha_2 = 0, \quad \gamma_3 = d > 0, \quad M = W^r, \quad N = 3^d W^s, \quad r\theta > s, \tag{52}$$

with

$$\beta = \frac{rd\theta}{r\theta - s}, \qquad \theta - \frac{s}{r} = \frac{d\theta}{\beta}, \qquad \Delta = rsd > 0. \tag{53}$$

Family −:

$$\gamma_3 = 0, \quad \alpha_2 = d > 0, \quad M = 2^d W^r, \quad N = W^s, \quad s > r\theta, \tag{54}$$

with

$$\beta = \frac{sd}{s - r\theta}, \qquad 1 - \frac{r\theta}{s} = \frac{d}{\beta}, \qquad \Delta = -rsd < 0. \tag{55}$$

Proof. If $r = 0$, then Theorem 5.5(a) forces $\alpha_3 > 0$, contrary to (50); similarly $s = 0$ is impossible. Thus $r, s > 0$.

Leastness H1 says $\alpha_2 = 0$ or $\gamma_3 = 0$. Equation (51) and Theorem 5.2 say $\gamma_3 \neq \alpha_2$. Therefore exactly one of α_2, γ_3 is zero and the other is a positive integer d.

If $\alpha_2 = 0$ and $\gamma_3 = d$, then (35) gives $U = 0$ and $V = -rd$. Formula (37) becomes

$$\beta = \frac{-rd\theta}{s - r\theta} = \frac{rd\theta}{r\theta - s}.$$

Positivity of β forces $r\theta > s$, and rearrangement gives the rational approximation identity in (53). The value of Δ follows from (51).

The second case is symmetric: $U = sd$, $V = 0$, so $\beta = sd/(s - r\theta)$, positivity forces $s > r\theta$, and rearrangement gives (55). □

Corollary 5.10 (Localized radical equation)**.** *Both alternatives in Theorem 5.9 imply the single form*

$$\boxed{W^{|r\theta - s|} = 3^d,} \qquad W > 1, \quad \gcd(W, 6) = 1, \quad r, s, d \in \mathbb{N}. \tag{56}$$

Conversely, together with the appropriate sign of $r\theta - s$, the equation reconstructs the corresponding factorization in (52) *or* (54)*.*

Proof. In Family +, compare $N = 3^\beta$ with $N = 3^d W^s$ and use $M = W^r = 2^\beta$:

$$3^d W^s = 3^\beta = (2^\beta)^\theta = (W^r)^\theta,$$

so $W^{r\theta - s} = 3^d$. In Family −, compare $M = 2^\beta = 2^d W^r$ with $N = W^s = 3^\beta$; taking logarithms and using $\theta \log 2 = \log 3$ gives $(s - r\theta)\log W = d \log 3$. The converse is the same algebra in reverse. □

Remark 5.11 (A precise rational-approximation reading)**.** Family $+$ places s/r below θ, with exact relative error d/β. Family $-$ places $r\theta/s$ below 1, again with exact error d/β. These are not exceptionally strong approximations: when $r, s \asymp \beta$, the error is naturally of order $1/\beta$. Therefore a standard irrationality measure for θ does not eliminate the families. The useful new feature is the exact integer parameter $d = |\Delta|/(rs)$ and the localized radical identity (56).

5.6 The localized four-exponentials square

Equation (56) produces a second four-exponentials configuration, distinct from the original square with bases $2, 3$ and exponents $1, \beta$.

Theorem 5.12 (A symmetric localized square)**.** *Assume* (56) *and define*

$$\eta = \frac{\log 3}{\log W} = \frac{|r\theta - s|}{d} > 0. \tag{57}$$

Then

(i) $\log 2$ *and* $\log W$ *are linearly independent over* $\mathbb{Q}$*;*

(ii) θ *and* η *are linearly independent over* $\mathbb{Q}$*;*

(iii) all four numbers

$$2^\theta, \quad 2^\eta, \quad W^\theta, \quad W^\eta \tag{58}$$

are algebraic; in fact $2^\theta = W^\eta = 3$ *and the other two are explicit rational radicals.*

Thus a no-cross-prime rank-three counterexample gives an explicit 2×2 *instance forbidden by the Four Exponentials Conjecture.*

Proof. Because W is an integer coprime to 2 and greater than one, no nonzero integer powers of 2 and W coincide; hence their logarithms are $\mathbb{Q}$-independent.

Let $\sigma = \operatorname{sgn}(r\theta - s) \in \{+1, -1\}$. Then $d\eta = \sigma(r\theta - s)$. If $q\theta + q'\eta = 0$ with $q, q' \in \mathbb{Q}$, substitution gives

$$\left(q + \frac{\sigma r q'}{d}\right)\theta = \frac{\sigma s q'}{d}.$$

Since $s > 0$ and $\theta \notin \mathbb{Q}$, both rational coefficients must vanish; therefore $q' = q = 0$.

The diagonal identities are immediate. Moreover

$$2^\eta = \left(\frac{3^r}{2^s}\right)^{\sigma/d},$$

which is algebraic. If $\sigma = +1$, then $W^{r\theta - s} = 3^d$ gives

$$W^\theta = (3^d W^s)^{1/r};$$

if $\sigma = -1$, then $W^{s - r\theta} = 3^d$ gives

$$W^\theta = (W^s/3^d)^{1/r}.$$

Both are algebraic. ∎

Remark 5.13 (Why six exponentials still does not fire)**.** The localized square supplies two independent base logarithms and two independent exponents, exactly the open 2×2 threshold. Every additional base obtained multiplicatively from 2 and W has logarithm in the same two-dimensional $\mathbb{Q}$-space, so it cannot provide the third independent base needed by the Six Exponentials Theorem. The natural candidate 3 has 3^θ and 3^η uncontrolled. The reduction is therefore sharp rather than cosmetic.

6 A unified exterior–Möbius dichotomy

The two preceding sections look rather different. The rank-four branch is controlled by a family of local 2×2 valuation minors, whereas the rank-three branch is controlled by one global determinant Δ. They are in fact the two complementary outputs of the same exterior-algebra test.

Let

$$\mathcal{P}_{\mathrm{ext}} = \{p : p \geq 5 \text{ prime}\}, \qquad L_{\mathrm{ext}} = \bigoplus_{p \in \mathcal{P}_{\mathrm{ext}}} \mathbb{Z} e_p,$$

and encode the external valuations by

$$A = \sum_{p \geq 5} a_p e_p, \qquad B = \sum_{p \geq 5} b_p e_p. \tag{59}$$

Their exterior product is

$$\Omega(M, N) = A \wedge B = \sum_{p<q} (a_p b_q - a_q b_p) e_p \wedge e_q = \sum_{p<q} \omega_{pq} e_p \wedge e_q. \tag{60}$$

It is an integral, finitely supported bivector.

Theorem 6.1 (Exterior–Möbius dichotomy)**.** *For a hypothetical least nonintegral exponent β, exactly one of the following alternatives occurs.*

(R4) Exterior branch. *$\Omega(M, N) \neq 0$. Then the four positive algebraic numbers $2, 3, M, N$ have multiplicative rank four, and at least one valuation minor ω_{pq} is a nonzero integer. In the cross-coprime subbranch, Lemma 4.6 produces an orientation with $\omega_{pq} \geq 1$, and Theorem 4.10 gives the explicit energy gap.*

(R3) Möbius branch. *$\Omega(M, N) = 0$. Then the external valuation vectors lie on a unique primitive ray, so there are a primitive integer $W > 1$, coprime to 6, and $r, s \in \mathbb{Z}_{\geq 0}$, not both zero, such that*

$$M = 2^{\alpha_2} 3^{\alpha_3} W^r, \qquad N = 2^{\gamma_2} 3^{\gamma_3} W^s.$$

For these uniquely normalized data, the Möbius determinant $\Delta = -rU - sV$ of (39) is a nonzero integer. The four numbers $2, 3, M, N$ have multiplicative rank three.

Proof. By H4, the external valuation vectors are not both zero and their rational rank is either one or two. Their rank is two exactly when $A \wedge B \neq 0$, equivalently when at least one 2×2 minor ω_{pq} is nonzero. Since the structural primes 2 and 3 already supply two independent valuation directions, this is precisely the multiplicative-rank-four branch.

If $A \wedge B = 0$, the nonzero vectors A, B are rationally proportional. Because their coordinates are nonnegative integers, there is a unique primitive nonnegative integral vector $C = \sum c_p e_p$ and unique $r, s \in \mathbb{Z}_{\geq 0}$, not both zero, such that $A = rC$ and $B = sC$. Put $W = \prod p^{c_p}$. This gives the displayed rank-three factorization and makes W primitive. Proposition 5.2 then gives $\Delta \neq 0$. The alternatives are mutually exclusive because Ω cannot be both zero and nonzero. ☐

Definition 6.2 (Discrete branch certificates)**.** In branch (R4), define

$$I_4 = \gcd_{p<q} |\omega_{pq}|, \tag{61}$$

where zero minors do not affect the gcd. In branch (R3), define

$$I_3 = |\Delta|. \tag{62}$$

Then the active branch certificate is always a positive integer.

The certificate I_4 is local-to-global: it is assembled from pairs of external primes. The certificate I_3 is global-to-local: after all external minors vanish, it measures how the structural $(2, 3)$ directions twist the single external ray. This is why the two invariants do not compete. The second becomes visible only after the first has vanished.

hypothetical least counterexample with external valuation vectors A, B

$\Downarrow$ test $\Omega = A \wedge B$ $\Downarrow$

$\Omega \neq 0$: **rank four**
some $\omega_{pq} \in \mathbb{Z} \setminus \{0\}$
cross-coprime: $\mathcal{E} > 0$ and an explicit gap

$\Omega = 0$: **rank three**
primitive external core W
$\Delta = \det B_{\mathrm{Mob}} \in \mathbb{Z} \setminus \{0\}$

Table 2: Complementary arithmetic data in the two multiplicative-rank branches.

	rank four	**rank three**
External geometry	two-dimensional; $A \wedge B \neq 0$	one-dimensional; $A \wedge B = 0$
Discrete witness	a local minor $\omega_{pq} = a_p b_q - a_q b_p$	the global Möbius determinant $\Delta = -rU - sV$
Integrality gap	$\lvert\omega_{pq}\rvert \geq 1$	$\lvert\Delta\rvert \geq 1$
Analytic avatar	normalized area $u_p v_q - u_q v_p$ and energy $\mathcal{E}$	nondegenerate conic $\Phi(X, Y, Z) = 0$ with $\det S = -\Delta/4$
Four-exponentials wall	the original square $(2, 3;\ M, N)$	a localized square $(2, W;\ \theta, \eta)$ in the no-cross-prime subbranch
Current missing input	wedge-sensitive cancellation or a stronger determinant	exclusion of the corresponding integral special radical

6.1 An exact one-radical reduction of the rank-three branch

The conic identity can be solved for the normalized logarithm of the primitive core. This gives a compact arithmetic target that is useful both for proof search and for formalization.

Let

$$\rho_* = \frac{\gamma_2 + (\gamma_3 - \alpha_2)\theta - \alpha_3\theta^2}{r\theta - s}, \qquad \beta_* = \alpha_2 + \alpha_3\theta + r\rho_*. \tag{63}$$

The denominator is nonzero by (36).

Theorem 6.3 (One-radical rank-three reduction)**.** *Fix nonnegative integers $\alpha_2, \alpha_3, \gamma_2, \gamma_3, r, s$ with $(r, s) \neq (0, 0)$, $r\theta \neq s$, and $\Delta \neq 0$. Assume $\rho_* > 0$ and $\beta_* > 0$. Then the following are equivalent.*

(i) There is an integer $W > 1$, coprime to 6, such that

$$2^{\beta_*} = 2^{\alpha_2} 3^{\alpha_3} W^r, \qquad 3^{\beta_*} = 2^{\gamma_2} 3^{\gamma_3} W^s.$$

(ii) The single special radical

$$W_* = 2^{\rho_*} = 2^{\frac{\gamma_2 + (\gamma_3 - \alpha_2)\theta - \alpha_3\theta^2}{r\theta - s}} \tag{64}$$

is an integer > 1 coprime to 6.

When these conditions hold, $W = W_*$ and $\beta = \beta_*$.

Proof. If (i) holds, take base-2 logarithms and put $\rho = \log W/\log 2$. Eliminating β from

$$\beta = \alpha_2 + \alpha_3\theta + r\rho, \qquad \beta\theta = \gamma_2 + \gamma_3\theta + s\rho$$

gives $\rho = \rho_*$, so $W = 2^{\rho_*}$.

Conversely, assume (ii) and set $W = W_*$. The definition of β_* gives

$$2^{\beta_*} = 2^{\alpha_2}3^{\alpha_3}W^r.$$

The defining equation for ρ_* is exactly

$$(r\theta - s)\rho_* = \gamma_2 + (\gamma_3 - \alpha_2)\theta - \alpha_3\theta^2,$$

which rearranges to

$$\beta_*\theta = \gamma_2 + \gamma_3\theta + s\rho_*.$$

Exponentiating base 2 gives the second equality. □

Remark 6.4. Theorem 6.3 does not solve the branch: deciding whether (64) can be an integer is another form of the same transcendence obstruction. Its value is precision. It removes M, N, W, and β as independent unknowns and isolates one explicitly parameterized radical whose integrality is necessary and sufficient. In the no-cross-prime families it collapses to $W^{|r\theta - s|} = 3^d$.

7 Closure attempts and what they actually require

The new invariants sharpen both branches, but neither is a concealed proof of the conjecture. This section records the strongest consequences that can be extracted without importing an unproved transcendence statement, and it states branch-specific finishing theorems precisely enough to guide the next round of work.

7.1 Why the energy gap does not close rank four

Let D_T denote a nonzero integer determinant in the mixed-interpolation construction of H3. Write its full guaranteed common divisor as

$$Q_T^{\mathrm{act}} = Q_T^{\mathrm{trop}} C_T, \qquad C_T \in \mathbb{Z}_{\geq 1}, \tag{65}$$

where C_T is the gain from cancellation among terms that have the same minimal valuation. The coarse archimedean estimate in the inherited method has the form

$$\limsup_{T\to\infty} \frac{\log |D_T|}{H_T} \leq 1. \tag{66}$$

A contradiction would follow if the divisor exponent exceeded the right-hand side. By Theorem 4.5, a sufficient cancellation theorem is therefore

$$\liminf_{T\to\infty} \frac{\log C_T}{H_T} > \frac{3}{10} + \frac{\mathcal{E}(M,N)}{20}. \tag{67}$$

The energy correction thus raises the cancellation tax. This is not a defect in the calculation: a factorization that is more visibly rank four creates more primewise disagreement and lowers the termwise minimum. Any successful rank-four determinant must exploit the same disagreement a second time, through signed cancellation, an exterior-algebraic alternation, or a sharper archimedean norm.

More generally, if a new determinant family has archimedean exponent A_∞ and tropical divisor exponent $7/10 - \mathcal{E}/20$, it suffices to prove

$$\liminf \frac{\log C_T}{H_T} > A_\infty - \frac{7}{10} + \frac{\mathcal{E}}{20}. \tag{68}$$

This formula is a useful accounting identity: every proposed determinant can be evaluated by the three numbers A_∞, $\mathcal{E}$, and the cancellation gain.

7.2 Why the integer minor gives no uniform analytic gap

The minor bound is discrete, but normalization divides it by β^2:

$$u_p v_q - u_q v_p = \frac{\log p \log q}{\beta^2 \log 2 \log 3}\omega_{pq}.$$

Consequently the forced energy separation is of order β^{-4}. Since a hypothetical least exponent is not known to have an absolute upper bound, this does not produce a uniform improvement over 7/10. In particular, compactness arguments on the probability simplex cannot replace the integer minor: the arithmetic points may approach the diagonal as the valuation heights grow.

This also explains why replacing the Gini norm by a smooth strictly convex norm is unlikely to be enough. Strict convexity detects noncollinearity, but without a height-sensitive arithmetic lower bound it supplies only an unspecified positive defect. The normalized minor provides the correct height scale and shows exactly how small that defect is allowed to be.

7.3 Why $|\Delta| \geq 1$ does not close rank three

The rank-three certificate gives

$$1 \leq |\Delta| \ll \beta^3.$$

There is no conflict between these inequalities. To turn integrality into a contradiction one would need an independent estimate $|\Delta| < 1$, or a nonzero algebraic quantity divisible by Δ whose archimedean size tends to zero. The present conic identity supplies neither: Δ is a coefficient determinant, not the value of a small linear form.

The same issue appears in Möbius language. Equation (37) places β in the integral PGL_2-orbit of θ, with determinant $-\Delta$. Integral Möbius transformations of arbitrarily large determinant are plentiful. The missing theorem must use the simultaneous integrality of 2^β and 3^β, not merely the orbit relation.

7.4 Classical linear forms and exponentials theorems

Baker's theorem is decisive for a vanishing *linear* form in logarithms of algebraic numbers with algebraic coefficients. The rank-three relation (44) is instead quadratic in θ, or equivalently bilinear in the three logarithms X, Y, Z. When $\Delta = 0$ the quadratic factors and Baker-type rank reduction applies; Proposition 5.2 shows that a counterexample is forced into the complementary nonfactorable case. This is a structural explanation for the repeated Baker dead end in the unified report.

The six exponentials theorem also stops one step short. In the localized square (58), the base logarithms $\log 2$ and $\log W$ span only a two-dimensional $\mathbb{Q}$-space. The theorem needs three independent base logarithms against two independent exponents. Adjoining a third algebraic base guarantees that at least one of six values is transcendental, but does not force the four already-known entries in the 2×2 square to contain a transcendental number.

The matrix coefficient conjecture is designed precisely for matrices of logarithms at this rank boundary. The current Dasgupta–Kakde framework gives substantial structural results but does not, in the form presently available, prove the restricted coefficient statement needed here [5]. Thus invoking that framework without an additional special-case argument would only rename the remaining gap.

7.5 Branch-specific finishing theorems

The dichotomy suggests two sharply different targets.

Target statement 7.1 (Rank-four wedge-cancellation target)**.** *For the mixed-interpolation determinant attached to a cross-coprime output pair, the cancellation factor in* (65) *satisfies*

$$\liminf_{T\to\infty} \frac{\log C_T}{H_T} > \frac{3}{10} + \frac{\mathcal{E}(M,N)}{20}. \tag{69}$$

More ambitiously, the excess should admit a positive lower bound depending on one normalized nonzero exterior minor w_{pq}.

A proof of Conjecture 7.1, together with nonvanishing of the determinant, would eliminate the cross-coprime rank-four branch. The theorem should not be sought in termwise valuations; those have already been optimized by H3. It must analyze the initial form in the graded p-adic algebra and show that the coefficients indexed by a nonzero wedge do not all cancel at the tropical level.

Target statement 7.2 (Rank-three special-radical target)**.** *For every nonnegative integral sextuple $(\alpha_2, \alpha_3, \gamma_2, \gamma_3, r, s)$ satisfying the inherited leastness conditions, $(r, s) \neq (0, 0)$, $r\theta \neq s$, and $\Delta \neq 0$, the number in* (64) *is not an integer > 1 coprime to 6 whenever the associated ρ_*, β_* are positive.*

By Theorem 6.3, this target eliminates the whole rank-three branch, not merely the no-cross-prime subbranch. Its simplest visible special case is the following.

Target statement 7.3 (Localized no-cross-prime radical target)**.** *Let r, s, d be positive integers and let $W > 1$ be an integer coprime to 6. Then*

$$W^{|r\theta - s|} \neq 3^d. \tag{70}$$

Conjecture 7.3 is equivalent to excluding the two families of Proposition 5.9. It is stronger than merely asserting that θ is irrational, but substantially more specialized than the full four exponentials conjecture: one diagonal of the associated exponential square is fixed to the rational integer 3, while the second base is constrained to be an integer coprime to 6.

Status after the new analysis. The output space is now partitioned by two mutually exclusive positive integer certificates. Rank four has an exact probability-energy loss and a quantitative β^{-4} separation. Rank three has an exact integral Möbius discriminant, a nondegenerate formal conic, and a necessary-and- sufficient one-radical formulation. A complete proof still requires one new input of the kind stated in Conjecture 7.1 or Conjecture 7.2; neither follows from the classical theorems presently in the ledger.

8 A Lean formalization blueprint

The new arguments are unusually well suited to formalization. Their hard inputs are elementary: finite sums, unique factorization, inequalities for nonnegative reals, 2×2 determinants, and polynomial identities. The transcendence barrier enters only at the final explicitly marked targets. This section gives a dependency-aware Lean 4 plan. The declarations below are specifications rather than a claim that the code has already been accepted by Lean.

8.1 Representation choices

The external valuation vectors should be represented by finitely supported functions

$$a, b : \{p : \mathbb{N} \mid p \text{ prime},\ p \geq 5\} \to_0 \mathbb{N}.$$

In Lean this can be implemented either by `Finsupp` on the subtype of external primes or, during the analytic development, by an arbitrary finite index type. The latter makes the probability and energy lemmas reusable and isolates all number-theoretic bookkeeping in a thin adapter layer.

The recommended split is:

L1. prove the energy inequalities for a finite type ι and two probability vectors $u, v : \iota \to \mathbb{R}$;

L2. prove the straddling lemma for a finite weighted family of nonzero real numbers with weighted sum zero;

L3. instantiate ι by the finite union of the prime supports of M and N;

L4. prove the valuation-minor identities in `\Z` , coercing to `\R` only at the final normalized-wedge step;

L5. formalize rank three in a separate record with six natural-number parameters and one positive real θ.

This organization avoids mixing `Nat.factorization`, real logarithms, and finite-probability algebra in every proof.

8.2 Proposed file layout

Table 3: Suggested Lean module decomposition.

module	responsibility
`AlaogluErdos/Theta.lean`	define $\theta = \log 3/\log 2$; prove positivity, irrationality, and $\theta \log 2 = \log 3$
`AlaogluErdos/FiniteProbability.lean`	probability vectors, total variation, diagonal energy, and Lemma 4.8
`AlaogluErdos/GiniQuadrant.lean`	closed first-quadrant Gini formula and the exact energy decomposition
`AlaogluErdos/PrimeSlope.lean`	weighted-zero-sum straddling, local nonvanishing, and positivity of an integral minor
`AlaogluErdos/RankFourGap.lean`	normalized wedge, the constant κ, and Theorem 4.10
`AlaogluErdos/ExternalRay.lean`	primitive-ray decomposition of proportional nonnegative `Finsupp` vectors
`AlaogluErdos/MobiusData.lean`	U, V, Δ, Möbius formula, determinant identity, and zero patterns
`AlaogluErdos/FormalConic.lean`	polynomial Φ, symmetric matrix S, and $\det S = -\Delta/4$
`AlaogluErdos/SpecialRadical.lean`	Theorem 6.3 and no-cross-prime specialization
`AlaogluErdos/BranchDichotomy.lean`	exterior bivector and Theorem 6.1

8.3 Abstract probability layer

A practical definition avoids division by zero explicitly.

Listing 1: Lean-style specification of the energy and total variation.

```
def diagTerm (x y : Real) : Real :=
  if max x y = 0 then 0 else (x - y)^2 / max x y

def diagEnergy [Fintype ι] (u v : ι → Real) : Real :=
  ∑ i, diagTerm (u i) (v i)

def totalVariation [Fintype ι] (u v : ι → Real) : Real :=
  (1 / 2 : Real) * ∑ i, |u i - v i|
```

```
structure ProbVec (ι : Type*) [Fintype ι] where
  val    : ι → Real
  nonneg : ∀ i, 0 ≤ val i
  sum_eq : ∑ i, val i = 1
```

The central abstract theorem should be stated independently of primes.

Listing 2: Target statement for the energy–variation comparison.

```
theorem energy_tv_bounds [Fintype ι]
    (u v : ProbVec ι) :
    4 * totalVariation u.val v.val ^ 2 /
        (1 + totalVariation u.val v.val)
      ≤ diagEnergy u.val v.val
    ∧ diagEnergy u.val v.val
      ≤ 2 * totalVariation u.val v.val := by
  -- Set d i := u i - v i and m i := max (u i) (v i).
  -- Prove ∑ i, m i = 1 + TV.
  -- Apply finite Cauchy--Schwarz to |d i| / sqrt (m i).
  -- Use |d i| ≤ m i for the upper bound.
  ...
```

A robust formal proof can avoid square roots and divisions by invoking the Engel form of Cauchy–Schwarz on the support where $m_i > 0$. Alternatively, prove

$$\left(\sum |d_i|\right)^2 \leq \left(\sum d_i^2/m_i\right)\left(\sum m_i\right)$$

by `Finset.sum_mul_sq_le_sq_mul_sq` after defining zero-safe terms.

The first-quadrant Gini formula should be formalized algebraically rather than through measure integration, because H3 already supplies the closed formula.

Listing 3: Algebraic first-quadrant identity.

```
theorem gini_firstQuadrant (u v : Real) (hu : 0 ≤ u) (hv : 0 ≤ v) :
    gini u v = (2 / 15 : Real) *
      (u + v + (u - v)^2 / max u v) := by
  rcases le_total u v with huv | hvu
  · rw [gini_closed_of_nonneg_of_le hu huv]
    field_simp [max_eq_right huv]
    ring
  · rw [gini_symm, gini_closed_of_nonneg_of_le hv hvu]
    field_simp [max_eq_left hvu]
    ring
```

The actual implementation needs a separate origin case before `field_simp`. This is a routine but important guard: an automated proof that silently assumes `max u v \ne 0` would formalize a stronger and false statement.

8.4 Straddling and the integer minor

The sign argument has no number theory in it.

Listing 4: Abstract weighted straddling lemma.

```
theorem exists_pos_and_neg_of_weighted_sum_zero
    [Fintype ι] [Nonempty ι]
    (w δ : ι → Real)
    (hw : ∀ i, 0 < w i)
    (hδ : ∀ i, δ i ≠ 0)
    (hsum : ∑ i, w i * δ i = 0) :
    (∃ p, 0 < δ p) ∧ (∃ q, δ q < 0) := by
  constructor
```

```
  · by_contra h
    push_neg at h
    have : ∑ i, w i * δ i < 0 := by
      -- every δ i is strictly negative
      ...
    linarith
  · by_contra h
    push_neg at h
    have : 0 < ∑ i, w i * δ i := by
      -- every δ i is strictly positive
      ...
    linarith
```

The local nonvanishing lemma can be based on irrationality of θ.

Listing 5: Local slope nonvanishing and minor positivity.

```
theorem nat_mul_theta_ne_nat
    (hθ : Irrational θ) (a b : Nat) (hab : a ≠ 0 ∨ b ≠ 0) :
    (a : Real) * θ - b ≠ 0 := by
  intro h
  have ha : a ≠ 0 := by
    intro ha0
    simp [ha0] at h
    exact hab.resolve_right (Nat.cast_eq_zero.mp h)
  have : θ = (b : Real) / a := by
    field_simp [ha]
    linarith
  exact hθ.ne_rat this

theorem minor_pos_of_straddles
    (aP bP aQ bQ : Nat)
    (hp : 0 < (aP : Real) * θ - bP)
    (hq : (aQ : Real) * θ - bQ < 0) :
    0 < (aP * bQ : Int) - aQ * bP := by
  -- Rewrite the real coercion as
  -- aP*(bQ-aQ*θ) + aQ*(aP*θ-bP), then use integrality.
  ...
```

After this theorem, the lower bound $\omega_{pq} \geq 1$ is immediate from `Int.add_one_le_iff` or `Int.one_le_iff_ne_zero` plus positivity. The normalized wedge identity is a single `ring` proof once the definitions of u_p and v_p are unfolded.

The inequality $w_{pq} \leq \mathrm{TV}$ is again abstract and should be proved for any pair of probability vectors. This keeps all real-analysis details out of the valuation module.

8.5 Rank-three data as an integral record

Subtraction makes `Nat` cumbersome, so the derived quantities U, V, Δ should live in `Int` from the outset.

Listing 6: Proposed rank-three parameter record.

```
structure RankThreeData where
  a2 a3 g2 g3 r s : Nat
  rs_ne_zero : r ≠ 0 ∨ s ≠ 0

def RankThreeData.U (d : RankThreeData) : Int :=
  d.s * d.a2 - d.r * d.g2

def RankThreeData.V (d : RankThreeData) : Int :=
  d.s * d.a3 - d.r * d.g3

def RankThreeData.Delta (d : RankThreeData) : Int :=
```

```
  -(d.r : Int) * d.U - (d.s : Int) * d.V
```

The first key identity is purely integral.

Listing 7: Expanded discriminant identity.

```
theorem delta_expanded (d : RankThreeData) :
    d.Delta =
      (d.r : Int)^2 * d.g2
      + (d.r : Int) * d.s * (d.g3 - d.a2)
      - (d.s : Int)^2 * d.a3 := by
  simp [RankThreeData.Delta, RankThreeData.U, RankThreeData.V]
  ring
```

The Möbius determinant can use `Matrix (Fin 2) (Fin 2) Int`; for a 2×2 matrix, it may be simpler to prove the explicit determinant formula than to unfold the generic permutation definition repeatedly.

Listing 8: Möbius determinant identity.

```
def mobiusMatrix (d : RankThreeData) : Matrix (Fin 2) (Fin 2) Int :=
  !![d.V, d.U; -(d.r : Int), d.s]

theorem det_mobiusMatrix (d : RankThreeData) :
    Matrix.det (mobiusMatrix d) = -d.Delta := by
  simp [mobiusMatrix, RankThreeData.Delta]
  ring
```

The implication $\Delta = 0 \Rightarrow \beta \in \mathbb{Q}$ should be split by the zero patterns $r = 0$, $s = 0$, and $r, s > 0$, exactly as in Proposition 5.2. This is longer than the paper proof but avoids hidden denominator assumptions. The final contradiction can use the already formalized rational-exponent lemma:

$$x \in \mathbb{Q}, \quad 2^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z}.$$

8.6 Formal quadratic bridge

The identity $\det S = -\Delta/4$ is ideal for `ring_nf`. To avoid fractions, formalize the doubled matrix

$$2S = \begin{pmatrix} 2\gamma_2 & \gamma_3 - \alpha_2 & s \\ \gamma_3 - \alpha_2 & -2\alpha_3 & -r \\ s & -r & 0 \end{pmatrix}, \tag{71}$$

for which

$$\det(2S) = -2\Delta. \tag{72}$$

This stays entirely in `Int`; the rational identity follows by coercion.

Listing 9: Integer version of the formal-conic determinant.

```
def doubledConicMatrix (d : RankThreeData) : Matrix (Fin 3) (Fin 3) Int :=
  !![2*d.g2, d.g3-d.a2, d.s;
     d.g3-d.a2, -2*d.a3, -d.r;
     d.s, -d.r, 0]

theorem det_doubledConicMatrix (d : RankThreeData) :
    Matrix.det (doubledConicMatrix d) = -2 * d.Delta := by
  simp [doubledConicMatrix, RankThreeData.Delta,
        RankThreeData.U, RankThreeData.V, Matrix.det_fin_three]
  ring
```

The conic identity itself is best stated in a commutative ring, not only in $\mathbb{R}$. That makes explicit that no analytic property of logarithms is used in the bridge.

8.7 Special-radical equivalence at the logarithmic level

Formalizing `Real.rpow` too early creates avoidable side conditions. First prove an equivalence between two affine equations in β, ρ and the single formula for ρ_*. Only then instantiate $\rho = \log W / \log 2$ and exponentiate.

Listing 10: Linear-algebra core of the special radical reduction.

```
theorem rho_eq_iff_two_log_equations
    (d : RankThreeData) (hden : (d.r : Real) * θ ≠ d.s)
    (rho beta : Real) :
    (beta = d.a2 + d.a3 * θ + d.r * rho ∧
     beta * θ = d.g2 + d.g3 * θ + d.s * rho) ↔
    (rho =
      (d.g2 + (d.g3-d.a2)*θ - d.a3*θ^2) /
      ((d.r : Real)*θ-d.s) ∧
     beta = d.a2 + d.a3*θ + d.r*rho) := by
  constructor <;> intro h
  · rcases h with ⟨h1,h2⟩
    constructor
    · field_simp [hden]
      nlinarith
    · exact h1
  · rcases h with ⟨hr,hb⟩
    constructor
    · exact hb
    · field_simp [hden] at hr
      nlinarith
```

The final exponential statements use the standard identities

$$\texttt{Real.rpow\_add}, \qquad \texttt{Real.rpow\_natCast}, \qquad 2^{\theta} = 3.$$

Because all bases are positive, no branch-cut issue occurs.

8.8 Dichotomy and dependency graph

The exterior dichotomy is a finite-rank statement about `Finsupp` vectors. Mathlib already provides exterior algebra, but importing it is unnecessary for this theorem: define `allMinorsZero a b` and prove directly that all minors vanish iff the two nonnegative integral vectors lie on a common primitive ray. The actual bivector can be added later as a conceptual wrapper.

rank-four dependency chain	**rank-three dependency chain**
`Theta` ⟶ `PrimeSlope`	`Theta` ⟶ `MobiusData`
`FiniteProb.` ⟶ `RankFourGap`	`ExternalRay` ⟶ `MobiusData`
`PrimeSlope` ⟶ `RankFourGap`	`MobiusData` ⟶ `FormalConic`
`RankFourGap` ↘	`MobiusData` ⟶ `SpecialRadical`
⟶ `BranchDichotomy` ⟵	

A useful formalization milestone is the following theorem with all analytic hypotheses abstracted away:

> Given two nonnegative integral external valuation vectors of positive total logarithmic masses satisfying the slope-conservation identity, either a positive integral minor exists and forces the normalized energy bound, or the vectors have a primitive common ray and the associated Möbius determinant is nonzero whenever the induced exponent is irrational.

This theorem captures every new unconditional argument in the report. The two finishing conjectures should remain separate axioms or theorem holes; hiding either inside a broad transcendence import would defeat the diagnostic value of the formalization.

9 Symbolic, numerical, and logical audit

The formulas in this report were checked in three complementary ways: hand derivation, exact symbolic expansion, and limiting-case tests. This section records enough detail for independent reproduction.

9.1 Exact polynomial identities

The following compact SymPy session verifies the three identities most prone to sign or factor errors.

Listing 11: Exact symbolic checks used in the report.

```
import sympy as s

u, v = s.symbols('u v', positive=True)
g0 = 2*(2*u**2 + 3*u*(v-u) + 2*(v-u)**2)/(15*v)
g1 = s.Rational(2,15)*(u+v+(u-v)**2/v)
assert s.simplify(g0-g1) == 0          # first-quadrant Gini

a2,a3,g2,g3,r,t = s.symbols('a2 a3 g2 g3 r s')
Delta = r**2*g2 + r*t*(g3-a2) - t**2*a3
S = s.Matrix([
 [g2,       (g3-a2)/2,  t/2],
 [(g3-a2)/2,   -a3,    -r/2],
 [t/2,         -r/2,     0]])
assert s.factor(S.det()) == -Delta/4   # formal conic

U = t*a2-r*g2
V = t*a3-r*g3
B = s.Matrix([[V,U],[-r,t]])
assert s.factor(B.det()) == -Delta     # Mobius determinant
```

The doubled integral matrix (71) provides an independent factor check: because a 3×3 determinant is cubic under scalar multiplication,

$$\det(2S) = 2^3 \det S = 8\left(-\frac{\Delta}{4}\right) = -2\Delta.$$

9.2 Numerical constants

Using natural logarithms,

$$\kappa = \frac{\log 5 \log 7}{\log 2 \log 3} = 4.1127006240\ldots, \tag{73}$$

$$\frac{\kappa^2}{5} = 3.3828612847\ldots, \tag{74}$$

$$\frac{\kappa^2}{10} = 1.6914306423\ldots. \tag{75}$$

The coefficient in (29) is the last of these numbers. The constants are dimensionless and independent of the logarithm base.

The cubic discriminant bound uses

$$\frac{3 \log 2 \log 3}{(\log 5)^2} = 0.8809\ldots. \tag{76}$$

9.3 Limiting cases for the energy

Three test configurations expose the normalization and all endpoint factors. They are tests of the analytic identity, not claims that the configurations come from counterexamples.

Coincident distributions. If $u = v$, then $\mathcal{E} = 0$ and (11) gives $7/10$. Theorem 4.5 shows that arithmetic factorizations cannot realize this endpoint.

Disjoint distributions. For $u = (1, 0)$ and $v = (0, 1)$, one has $\mathrm{TV} = 1$ and $\mathcal{E} = 2$; hence the tropical constant is $7/10 - 2/20 = 3/5$. Both inequalities in the allowed range are therefore sharp on the abstract probability simplex.

Symmetric near-diagonal perturbation. For $0 < \varepsilon < 1/2$, let

$$u = \left(\frac{1}{2} + \varepsilon, \frac{1}{2} - \varepsilon\right), \qquad v = \left(\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon\right).$$

Then

$$\mathrm{TV} = 2\varepsilon, \qquad \mathcal{E} = \frac{16\varepsilon^2}{1 + 2\varepsilon}. \tag{77}$$

Thus the lower inequality in Lemma 4.8 is attained exactly. This verifies both the factor 4 and the denominator $1 + \mathrm{TV}$ and shows that the quadratic small-defect scale is optimal.

9.4 Sign tests for the Möbius discriminant

The axis cases provide immediate sign checks:

$$r = 0 < s \Longrightarrow \Delta = -s^2\alpha_3, \qquad s = 0 < r \Longrightarrow \Delta = r^2\gamma_2.$$

These agree with both $-rU - sV$ and the expanded quadratic formula.

In no-cross family $(+)$, substituting $\alpha_2 = \alpha_3 = \gamma_2 = 0$, $\gamma_3 = d$ gives $\Delta = rsd > 0$. In family $(-)$, substituting $\alpha_3 = \gamma_2 = \gamma_3 = 0$, $\alpha_2 = d$ gives $\Delta = -rsd < 0$. The signs agree with the direction of the approximation $r\theta - s$.

A useful synthetic unit-determinant pattern is

$$\alpha_2 = \alpha_3 = \gamma_3 = 0, \quad \gamma_2 = 1, \quad r = s = 1,$$

for which $\Delta = 1$ and

$$\rho_* = \beta_* = \frac{1}{\theta - 1}, \qquad W_* = 2^{1/(\theta-1)}.$$

Nothing elementary forces this special radical to be nonintegral. This test confirms that $|\Delta| = 1$ is a genuine boundary case rather than an immediate contradiction.

9.5 Dependency and noncircularity audit

The unconditional chain in rank four is

$$\text{factorizations} \Longrightarrow \text{probability vectors} \Longrightarrow \text{slope conservation} \Longrightarrow \omega_{pq} \geq 1,$$

$$\Longrightarrow \mathrm{TV} \geq \kappa/\beta^2 \Longrightarrow \mathcal{E} \geq \frac{4\kappa^2}{\beta^4 + \kappa\beta^2}.$$

No determinant nonvanishing conjecture or four-exponentials statement is used in this chain. H3 is used only in the final conversion from energy to the tropical divisor exponent.

The unconditional chain in rank three is

$$A \wedge B = 0 \Longrightarrow (W, r, s) \Longrightarrow \beta = \frac{U + V\theta}{s - r\theta} \Longrightarrow \Delta \neq 0 \Longrightarrow \det S = -\Delta/4 \Longrightarrow \text{special radical } W_*.$$

The proof that $\Delta \neq 0$ uses only the inherited rational-exponent lemma and the assumption that β is a counterexample. The localized four-exponentials square is a consequence, not an input.

The public model-challenge reports and current literature reconnaissance are contextual only. No mathematical step in the two chains depends on a claimed external breakthrough, unpublished formalization, or machine-generated proof.

10 Conclusions and research priorities

This continuation does not claim a complete proof of the Alaoglu–Erdős conjecture. It does produce a branch-exhaustive arithmetic refinement that was absent from the unified report.

In rank four, the normalized external factorizations are probability measures. The first-quadrant Gini norm splits exactly into total mass plus a max-χ^2 energy. Unique factorization excludes zero energy, and a signed prime-slope conservation law forces a positive integer valuation minor. That minor yields an explicit total-variation separation and the quantitative ceiling

$$\frac{7}{10} - \frac{\kappa^2}{5(\beta^4 + \kappa\beta^2)}.$$

The old strictness clause is therefore replaced by an exact formula and a height-sensitive arithmetic deficit.

In rank three, vanishing of every external minor produces a primitive core W. The remaining structural information is measured by

$$\Delta = -rU - sV = \det(-B),$$

a nonzero integer. The same number is, up to $-1/4$, the determinant of the formal quadratic relation among $\log 2, \log 3$, and $\log W$. It classifies the zero patterns, separates the cross-coprime branch, yields a cubic height bound, and turns the whole rank-three branch into the integrality question for one explicit special radical.

The most promising next work is correspondingly bifurcated.

P1. Compute initial forms of the mixed determinant. The relevant object is not the minimum valuation but the sum of determinant terms attaining that minimum. The integer wedge ω_{pq} should be inserted into this graded calculation from the start. The concrete target is (69).

P2. Search for a rank-three special-radical theorem. Work directly with (64), exploiting nonnegativity, the leastness condition $\min(\alpha_2, \gamma_3) = 0$, coprimality of W with 6, and $|\Delta| \geq 1$. These restrictions are invisible in a generic four-exponentials formulation and may permit a specialized S-unit, modular, or p-adic argument.

P3. Formalize the unconditional core. The energy inequality, straddling minor, Möbius determinant, and formal conic are small, independent Lean milestones. A formal development would make sign conventions and zero cases completely stable while leaving the two genuine transcendence holes exposed.

P4. Use computation only as a theorem-discovery tool. Enumerate bounded rank-three parameter sextuples, compute Δ and high-precision ρ_*, and test proximity of 2^{ρ_*} to integers. Such a search cannot prove nonintegrality, but exceptional families may reveal congruence obstructions or a missing descent invariant.

The main conceptual gain is that the two hard branches now fail for different, explicit reasons. Rank four lacks enough cancellation in an otherwise quantified determinant. Rank three lacks an integrality obstruction for an otherwise explicit Möbius special radical. Treating those as separate problems is more informative than another global reformulation of the original 2×2 exponential square.

A Compact theorem ledger

Table 4: New statements and their logical status.

statement	**status**	**content**
Proposition 4.2	unconditional	exact first-quadrant Gini formula with diagonal-defect term
Theorem 4.5	unconditional from H3	tropical exponent $7/10 - \mathcal{E}/20$ and exclusion of equality
Lemma 4.6	unconditional	opposite local slopes force a positive integral valuation minor
Theorem 4.10	unconditional from H1–H3	TV $\geq \kappa/\beta^2$ and explicit β^{-4} energy gap
Proposition 5.2	unconditional from H1–H4	rank-three Möbius determinant is a nonzero integer
Proposition 5.4	unconditional	conic matrix determinant is $-\Delta/4$
Proposition 5.9	unconditional from H1–H4	two rank-three no-cross-prime families and $W^{\lvert r\theta - s\rvert} = 3^d$
Theorem 5.12	unconditional	second explicit four-exponentials square in the rank-three no-cross branch
Theorem 6.1	unconditional from H4	exhaustive split by Ω; active integer certificate I_4 or I_3
Theorem 6.3	unconditional algebra	necessary-and-sufficient one-radical reduction of rank three
Conjecture 7.1	open target	sufficient cancellation theorem for the cross-coprime rank-four branch
Conjecture 7.2	open target	sufficient special-radical exclusion for all of rank three

B Formula sheet

For quick reference, the two branches can be summarized as follows.

Rank four

$$u_p = \frac{a_p \log p}{\beta \log 2}, \qquad v_p = \frac{b_p \log p}{\beta \log 3}, \qquad \sum u_p = \sum v_p = 1,$$

$$\mathcal{E} = \sum_p \frac{(u_p - v_p)^2}{\max(u_p, v_p)}, \qquad \mathrm{TV} = \frac{1}{2}\sum_p |u_p - v_p|,$$

$$\frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow \frac{7}{10} - \frac{\mathcal{E}}{20}, \qquad \omega_{pq} = a_p b_q - a_q b_p \in \mathbb{Z} \setminus \{0\},$$

$$u_p v_q - u_q v_p = \frac{\log p \log q}{\beta^2 \log 2 \log 3}\omega_{pq}, \qquad \kappa = \frac{\log 5 \log 7}{\log 2 \log 3},$$

$$\mathrm{TV} \geq \frac{\kappa}{\beta^2}, \qquad \mathcal{E} \geq \frac{4\kappa^2}{\beta^4 + \kappa\beta^2}.$$

Rank three

$$M = 2^{\alpha_2} 3^{\alpha_3} W^r, \qquad N = 2^{\gamma_2} 3^{\gamma_3} W^s,$$

$$U = s\alpha_2 - r\gamma_2, \qquad V = s\alpha_3 - r\gamma_3,$$

$$\beta = \frac{U + V\theta}{s - r\theta}, \qquad \Delta = -rU - sV \in \mathbb{Z} \setminus \{0\},$$

$$\Phi(X, Y, Z) = \gamma_2 X^2 + (\gamma_3 - \alpha_2)XY - \alpha_3 Y^2 + sXZ - rYZ,$$

$$\det S = -\frac{\Delta}{4}, \qquad 1 \leq |\Delta| \ll \beta^3,$$

$$\rho_* = \frac{\gamma_2 + (\gamma_3 - \alpha_2)\theta - \alpha_3\theta^2}{r\theta - s}, \qquad W_* = 2^{\rho_*}.$$

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